Trigonometry

Identities, Transformations and Inverse Trigonometry

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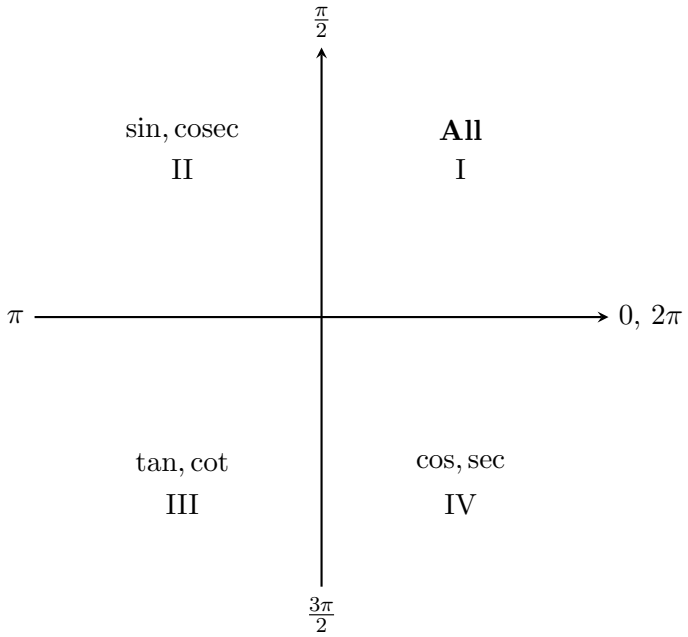
1 Trigonometric Identities

1.1 Standard Trigonometric Values

θ	$0°$	$30°$	$45°$	$60°$	$90°$
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Undefined
$\operatorname{cosec}\theta$	Undefined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Undefined
$\cot\theta$	Undefined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

1.2 ASTC Rule

The ASTC rule determines which trigonometric functions are positive in each quadrant.



1.3 Radians and Degrees

$$\text{Degrees} = \text{Radians} \times \frac{180°}{\pi}$$

$$\text{Radians} = \text{Degrees} \times \frac{\pi}{180°}$$

2 Negative Angles and Transformations

2.1 Negative Angle Identities

$$\sin(-x) = -\sin x$$
$$\cos(-x) = \cos x$$
$$\tan(-x) = -\tan x$$
$$\operatorname{cosec}(-x) = -\operatorname{cosec} x$$
$$\sec(-x) = \sec x$$
$$\cot(-x) = -\cot x$$

2.2 Transformation Rules

For angles involving $90°$ or $270°$, the trigonometric function changes to its cofunction.

Original	Transformed
sin	cos
cos	sin
tan	cot
cot	tan
sec	cosec
cosec	sec

For angles involving $180°$ or $360°$, the trigonometric function remains the same.

The sign is determined by the quadrant.

2.3 Common Transformations

$$\sin(90° - \theta) = \cos\theta$$
$$\cos(90° - \theta) = \sin\theta$$
$$\tan(90° - \theta) = \cot\theta$$
$$\cot(90° - \theta) = \tan\theta$$
$$\sec(90° - \theta) = \operatorname{cosec}\theta$$
$$\operatorname{cosec}(90° - \theta) = \sec\theta$$

$$\sin(90^\circ + \theta) = \cos\theta$$
$$\cos(90^\circ + \theta) = -\sin\theta$$
$$\tan(90^\circ + \theta) = -\cot\theta$$
$$\cot(90^\circ + \theta) = -\tan\theta$$
$$\sec(90^\circ + \theta) = -\operatorname{cosec}\theta$$
$$\operatorname{cosec}(90^\circ + \theta) = \sec\theta$$

$$\sin(180^\circ - \theta) = \sin\theta$$
$$\cos(180^\circ - \theta) = -\cos\theta$$
$$\tan(180^\circ - \theta) = -\tan\theta$$
$$\cot(180^\circ - \theta) = -\cot\theta$$
$$\sec(180^\circ - \theta) = -\sec\theta$$
$$\operatorname{cosec}(180^\circ - \theta) = \operatorname{cosec}\theta$$

$$\sin(180^\circ + \theta) = -\sin\theta$$
$$\cos(180^\circ + \theta) = -\cos\theta$$
$$\tan(180^\circ + \theta) = \tan\theta$$
$$\cot(180^\circ + \theta) = \cot\theta$$
$$\sec(180^\circ + \theta) = -\sec\theta$$
$$\operatorname{cosec}(180^\circ + \theta) = -\operatorname{cosec}\theta$$

$$\sin(270^\circ - \theta) = -\cos\theta$$
$$\cos(270^\circ - \theta) = -\sin\theta$$
$$\tan(270^\circ - \theta) = \cot\theta$$
$$\cot(270^\circ - \theta) = \tan\theta$$
$$\sec(270^\circ - \theta) = -\operatorname{cosec}\theta$$
$$\operatorname{cosec}(270^\circ - \theta) = -\sec\theta$$

$$\sin(270^\circ + \theta) = -\cos\theta$$
$$\cos(270^\circ + \theta) = \sin\theta$$
$$\tan(270^\circ + \theta) = -\cot\theta$$
$$\cot(270^\circ + \theta) = -\tan\theta$$
$$\sec(270^\circ + \theta) = \operatorname{cosec}\theta$$
$$\operatorname{cosec}(270^\circ + \theta) = -\sec\theta$$

$$\sin(360^\circ - \theta) = -\sin\theta$$
$$\cos(360^\circ - \theta) = \cos\theta$$
$$\tan(360^\circ - \theta) = -\tan\theta$$
$$\cot(360^\circ - \theta) = -\cot\theta$$
$$\sec(360^\circ - \theta) = \sec\theta$$
$$\operatorname{cosec}(360^\circ - \theta) = -\operatorname{cosec}\theta$$

$$\sin(360^\circ + \theta) = \sin\theta$$
$$\cos(360^\circ + \theta) = \cos\theta$$
$$\tan(360^\circ + \theta) = \tan\theta$$
$$\cot(360^\circ + \theta) = \cot\theta$$
$$\sec(360^\circ + \theta) = \sec\theta$$
$$\operatorname{cosec}(360^\circ + \theta) = \operatorname{cosec}\theta$$

2.4 Important Angle Conversions

$$\frac{\pi}{2} = 90^\circ \qquad \frac{\pi}{3} = 60^\circ$$
$$\frac{\pi}{4} = 45^\circ \qquad \frac{\pi}{6} = 30^\circ$$
$$\frac{3\pi}{2} = 270^\circ \qquad 2\pi = 360^\circ$$

3 Fundamental Trigonometric Identities

3.1 Pythagorean Identities

1. $\sin^2\theta + \cos^2\theta = 1$

2. $\operatorname{cosec}^2\theta - \cot^2\theta = 1$
3. $\sec^2\theta - \tan^2\theta = 1$

3.2 Compound Angle Identities

4. $\sin(x+y) = \sin x\cos y + \cos x\sin y$
5. $\sin(x-y) = \sin x\cos y - \cos x\sin y$
6. $\cos(x+y) = \cos x\cos y - \sin x\sin y$
7. $\cos(x-y) = \cos x\cos y + \sin x\sin y$
8. $$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x\tan y}$$
9. $$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x\tan y}$$
10. $$\cot(x+y) = \frac{\cot x\cot y - 1}{\cot x + \cot y}$$
11. $$\cot(x-y) = \frac{\cot x\cot y + 1}{\cot y - \cot x}$$

3.3 Double Angle Identities

12. $$\sin 2x = 2\sin x\cos x$$
13. $$\sin 2x = \frac{2\tan x}{1 + \tan^2 x}$$
14. $$\cos 2x = \cos^2 x - \sin^2 x$$
15. $$\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$
16. $$\cos 2x = 2\cos^2 x - 1$$
17. $$\cos 2x = 1 - 2\sin^2 x$$
18. $$\tan 2x = \frac{2\tan x}{1 - \tan^2 x}$$

4 Triple Angle and Product Formulas

4.1 Triple Angle Identities

19.
$$\sin 3x = 3\sin x - 4\sin^3 x$$

20.
$$\cos 3x = 4\cos^3 x - 3\cos x$$

21.
$$\tan 3x = \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x}$$

4.2 Sum to Product Identities

22.
$$\sin x + \sin y = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$

23.
$$\sin x - \sin y = 2\cos\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$$

24.
$$\cos x + \cos y = 2\cos\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$$

25.
$$\cos x - \cos y = -2\sin\left(\frac{x+y}{2}\right)\sin\left(\frac{x-y}{2}\right)$$

4.3 Product to Sum Identities

26.
$$2\sin x\cos y = \sin(x+y) + \sin(x-y)$$

27.
$$2\cos x\sin y = \sin(x+y) - \sin(x-y)$$

28.
$$2\cos x\cos y = \cos(x+y) + \cos(x-y)$$

29.
$$2\sin x\sin y = \cos(x-y) - \cos(x+y)$$

4.4 Half Angle Identities

30.
$$\sin\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{2}}$$

31.
$$\cos\frac{x}{2} = \pm\sqrt{\frac{1+\cos x}{2}}$$

32.
$$\tan\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{1+\cos x}}$$

Note: The sign depends on the quadrant in which $\frac{x}{2}$ lies.

5 Inverse Trigonometry

5.1 Regular Trigonometric Functions

$$\sin : \mathbb{R} \to [-1, 1]$$

$$\cos : \mathbb{R} \to [-1, 1]$$

$$\tan : \{x \in \mathbb{R} : x \neq \frac{(2n+1)\pi}{2},\ n \in \mathbb{Z}\} \to \mathbb{R}$$

$$\cot : \{x \in \mathbb{R} : x \neq n\pi,\ n \in \mathbb{Z}\} \to \mathbb{R}$$

$$\sec : \{x \in \mathbb{R} : x \neq \frac{(2n+1)\pi}{2},\ n \in \mathbb{Z}\} \to \{y \in \mathbb{R} : |y| \geq 1\}$$

$$\operatorname{cosec} : \{x \in \mathbb{R} : x \neq n\pi,\ n \in \mathbb{Z}\} \to \{y \in \mathbb{R} : |y| \geq 1\}$$

5.2 Inverse Trigonometric Functions

$$\sin^{-1} : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\cos^{-1} : [-1, 1] \to [0, \pi]$$

$$\tan^{-1} : \mathbb{R} \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$\cot^{-1} : \mathbb{R} \to (0, \pi)$$

$$\operatorname{cosec}^{-1} : \{x \in \mathbb{R} : |x| \geq 1\} \to \left\{y : -\frac{\pi}{2} \leq y \leq \frac{\pi}{2},\ y \neq 0\right\}$$

$$\sec^{-1} : \{x \in \mathbb{R} : |x| \geq 1\} \to \left\{y : 0 \leq y \leq \pi,\ y \neq \frac{\pi}{2}\right\}$$

6 Inverse Trigonometry Notes

6.1 Inverse Function vs Reciprocal

$$\boxed{\sin^{-1}(x) \neq (\sin x)^{-1}}$$

Here, $\sin^{-1} x$ represents the inverse sine function.

For example,

$$\sin 30^\circ = \frac{1}{2}$$

Therefore,

$$\sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$$

However,

$$(\sin x)^{-1} = \frac{1}{\sin x} = \operatorname{cosec} x$$

6.2 Trigonometric Substitutions

1. For expressions involving $a^2 + x^2$:
$$x = a\tan\theta \quad \text{or} \quad x = a\cot\theta$$
2. For expressions involving $1 + x^2$:
$$x = \tan\theta \quad \text{or} \quad x = \cot\theta$$
3. For expressions involving $a^2 - x^2$:
$$x = a\sin\theta \quad \text{or} \quad x = a\cos\theta$$
4. For expressions involving $1 - x^2$:
$$x = \sin\theta \quad \text{or} \quad x = \cos\theta$$
5. For expressions involving $x^2 - a^2$:
$$x = a\sec\theta \quad \text{or} \quad x = a\operatorname{cosec}\theta$$
6. For expressions involving $x^2 - 1$:
$$x = \sec\theta \quad \text{or} \quad x = \operatorname{cosec}\theta$$

6.3 Additional Substitutions

7. For expressions involving
$$\frac{1-x}{1+x} \quad \text{or} \quad \frac{1+x}{1-x},$$
use the substitution
$$x = \cos 2\theta.$$
8. For expressions involving
$$\sqrt{\frac{1-x}{1+x}} \quad \text{or} \quad \sqrt{\frac{1+x}{1-x}},$$
use
$$x = \cos 2\theta.$$

6.4 Additional Identities

9.
$$\cos x = \cos^2\left(\frac{x}{2}\right) - \sin^2\left(\frac{x}{2}\right)$$

10.
$$1 - \sin x = \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)^2$$

11.
$$1 + \sin x = \left(\cos\frac{x}{2} + \sin\frac{x}{2}\right)^2$$

7 Inverse Trigonometric Formulas

7.1 Reciprocal Relationships

1.
$$\sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x \qquad (|x| \geq 1)$$

2.
$$\cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x \qquad (|x| \geq 1)$$

3.
$$\tan^{-1}\left(\frac{1}{x}\right) = \cot^{-1} x \qquad (x > 0)$$

7.2 Complementary Inverse Identities

4.
$$\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$$

5.
$$\operatorname{cosec}^{-1} x + \sec^{-1} x = \frac{\pi}{2}$$

6.
$$\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$$

7.3 Inverse Tangent Addition and Subtraction

7.
$$\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

8.
$$\tan^{-1} x - \tan^{-1} y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$$

Note: These formulas require appropriate principal value and denominator conditions. Otherwise, an adjustment by π may be needed.

7.4 Double Inverse Tangent Formulas

10.
$$2\tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

11.
$$2\tan^{-1} x = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$$

12.
$$2\tan^{-1} x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$$

Note: These expressions are subject to principal value restrictions.

7.5 Negative Argument Identities

13.
$$\sin^{-1}(-x) = -\sin^{-1} x$$

14.
$$\cos^{-1}(-x) = \pi - \cos^{-1} x$$

15.
$$\tan^{-1}(-x) = -\tan^{-1} x$$

16.
$$\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$$

17.
$$\sec^{-1}(-x) = \pi - \sec^{-1} x$$

18.
$$\cot^{-1}(-x) = \pi - \cot^{-1} x$$